

Project Initialization and planning phase

Date	28 September 2026
Team ID	SWTID-2026-6259
Project Title	AI StudyBuddy API
Maximum Marks	5 Marks

Project Proposal (Proposal Solution) report:

The proposal aims to develop an AI-powered study assistant that helps learners understand and revise study materials efficiently. The system uses Generative AI to create summaries, flashcards, quizzes, and personalized study plans, reducing study time and improving learning support.

Project Overview:

Objective	The primary objective is to develop an AI-powered study assistant that provides quick and personalized learning support using Generative AI.
Scope	The project focuses on uploading study materials and generating summaries, flashcards, quizzes and personalized study plans through an easy-to-use web application.

Problem statement:

Description	Learners often face difficulty understanding lengthy study materials and preparing notes, quizzes, and revision resources manually.
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Impact	This results in increased study time, difficulty in revision, confusion, and stress during exam preparation.
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Proposed Solutions:

Approach	Developing an AI-powered learning platform using React.js, Node.js, Express.js, MongoDB, and Google Gemini AI to analyze study materials and generate useful learning resources.
Key Features	AI-generated study material summaries. - Flashcard generation for quick revision. - AI-generated quizzes. - Personalized study plans. - Secure user authentication and resource storage.

Resource Requirements:

Resource Type	Description	Specification/Allocation
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Hardware

Computing Resource	Processor for application development	Intel/AMD processor
Memory	RAM specification	8GB
Storage	Disk space for project files and database	256 GB SSD or above

Software

Frameworks	Frontend and backend frameworks	
Libraries	Required development libraries	
Development Environment	IDE/Development tools	

Data

Data	Data Storage for users, materials and generated resources	MongoDB with Mongoose
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